

Martel Vasquez

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION:

Bachelor of Science in Computer Science
George Mason University, College of Engineering & Computing
GPA: 3.70

Fairfax, VA
Cum Laude

TECHNICAL PROJECTS:

- **Django Note API (Python, Django, TastyPie):** Built a RESTful API for a note-taking backend. Designed models, configured routing, implemented CRUD operations, and tested endpoints using Postman to validate request/response behavior.
- **Set-UID Buffer Overflow Lab (C):** Exploited a vulnerable Set-UID C program to escalate privileges and spawn a root shell. Bypassed ASLR, NX, and StackGuard protections, demonstrating a strong understanding of system-level vulnerabilities and security testing.
- **Flextimer App (Dart/Flutter):** Created a workout timer app allowing users to save routines, target muscle groups, and receive recommendations. Contributed to requirements, advanced features, and UI design (light/dark mode, app info), using Agile workflow and unit testing.
- **Musical Chairs Simulation (Java):** Simulated a multithreaded game of musical chairs using Java concurrency tools. Applied synchronization and thread coordination to mimic real-world race conditions and resolve thread safety issues.
- **Cache Simulator (Java):** Developed a simulator implementing FIFO, LFU, and LRU cache replacement policies using linked lists and hash maps. Tested cache hit/miss logic to optimize data retrieval by 20%.
- **Reinforcement Learning Agents (Python, AI/ML):** Built value iteration, Q-learning, and approximate Q-learning agents for Pacman and Gridworld. Trained agents to learn optimal policies, achieving 90%+ win rate. Applied Bellman updates, exploration strategies, and feature-based generalization.
- **Customer Order Management CLI (Java, Oracle SQL):** Built a command-line interface that loads and interacts with SQL data using Oracle JDBC. Enabled users to display tables, search by attributes, and delete filtered results. Automated SQL script execution from user-provided file paths, streamlining data load and improving query flexibility by 30%.

CERTIFICATIONS:

- CompTIA Network+ (Expected September 2025)
- CompTIA Security+ (Expected November 2025)

SKILLS:

- Languages: Java, Python, C, SQL, HTML, CSS, JavaScript, Dart
- Frameworks & Dev Tools: IntelliJ, VS Code, Linux, Postman, Django, JUnit, Flutter
- Platforms: Windows, macOS
- Version Control: Git (GitHub, GitLab)

RELEVANT COURSEWORK:

- Data Structures, Software Engineering, Artificial Intelligence, Object-Oriented Programming, Formal Methods & Models, Computer Systems & Programming, Operating Systems, Analysis of Algorithms, Secure Programming and Systems, Database Concepts, Comparative Programming Languages

PROFESSIONAL EXPERIENCE:

Footwear Associate
Dicks Sporting Goods

August 2021 – Present
Fairfax, VA

- Delivered hands-on customer support and issue resolution in a fast-paced retail environment.
- Boosted sales by 20% through product knowledge and tailored recommendations.
- Recognized for professionalism, teamwork, and consistent customer satisfaction improvements.